

The Derivative

Definition:

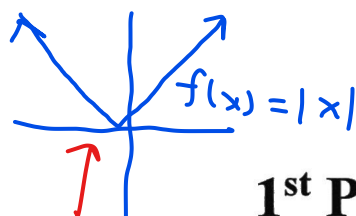
The derivative is the slope of the tangent line to the curve at any point x . It can also be defined as the rate of change at any point x .



The function must be defined at the point x .

Examples of where the function is **not** defined include: Vertical Asymptotes (V. A.) and holes.

Examples of where the function is **not** differentiable are sharp corners, endpoints, vertical tangents, **CUSP**



NO
unique tangent

1st Principles Definition of the Derivative

Lagrange Notation

read as "y prime"

$$m_t = y' = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

read as "f prime of x" or

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$y = 3x^3$$

$$y' = 27x^2$$

$$f(x) = 3x^3$$

$$f'(x) = 27x^2$$

$$f'(2) = 27(2)^2 = 108$$

Leibniz Notation

get the
derivative
of function
y w.r.t x.

read as "dy by dx"

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$



Example 1: Determine y' if $y = \sqrt{x+1}$.

$$y' = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$y' = \lim_{h \rightarrow 0} \frac{(\sqrt{x+h+1} - \sqrt{x+1})}{h} \cdot \frac{(\sqrt{x+h+1} + \sqrt{x+1})}{(\sqrt{x+h+1} + \sqrt{x+1})}$$

$$y' = \lim_{h \rightarrow 0} \frac{\cancel{x+h+1} - (\cancel{x+1})}{h(\sqrt{x+h+1} + \sqrt{x+1})}$$

$$y' = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h+1} + \sqrt{x+1}}$$

$$y' = \frac{1}{2\sqrt{x+1}}$$

$$y = 2x^2 + 6x + 11$$

Example 2: Determine $\frac{dy}{dx}$ at $x=3$ if $y = 2x^2 + 6x + 11$.

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(x+h)^2 + 6(x+h) + 11 - (2x^2 + 6x + 11)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(x^2 + 2xh + h^2) + 6x + 6h + 11 - 2x^2 - 6x - 11}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{2x^2} + 4xh + 2h^2 + \cancel{6x} + 6h + \cancel{11} - \cancel{2x^2} - \cancel{6x} - \cancel{11}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 + 6h}{h} \therefore \frac{dy}{dx} \bigg|_{x=3} = 18$$

$$= 4x + 6$$

$$\frac{dy}{dx} \bigg|_{x=3}$$

$$(3, 33)$$